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Superconductivity in High-Entropy Alloy System Containing Tb

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Abstract: Superconducting alloy containing terbium (Tb) and its reference without the lanthanide were synthesized. X-ray diffraction, scanning electron microscopy, energy dispersive X-ray spectroscopy, specific heat, and magnetic measurements were used to investigate their structural and physical properties. Both alloys crystallized in body-centered cubic structure, and the presence of small amounts of Tb and Tb_2O_3 phases was detected. The critical temperature T_c of alloys was in the 4.6–5.2 K range, and the upper critical field $\mu_0 H_{c2}$ was 6.1–6.8 T. The comparison with the reference determined the effect of Tb on the alloy's critical parameters and phase stability connected to the high-entropy alloys' core effects. Overall, Tb addition did not have a beneficial effect on the superconducting properties of this alloy.

Keywords: high-entropy alloys; superconductivity; terbium

1. Introduction

High-entropy alloys are among the most studied materials in modern solid-state physics and materials science. They can be defined as solid solutions of five or more elements with at least 5 atomic percent each. Their name derives from the high value of entropy of mixing, which is responsible for stabilizing simple structures (such as body-centered cubic, face-centered cubic, and hexagonal close-packed) in these multicomponent materials. This is because increasing the amount of elements in a material usually leads to the precipitation of binary and ternary phases. In recent years, these materials have attracted considerable attention because of their remarkable mechanical properties; resistance to corrosion, including chemical corrosion; and stability over a wide temperature range [1–4]. These features are due to the so-called HEA core effects described by Yeh et al. [5,6].

Therefore, another consequence of these effects is the strong influence of internal chemical disorder and crystal lattice stresses on the superconducting properties of selected HEA alloys. Superconducting HEAs form a steadily growing family currently dominated by Nb-Ta matrices [7–9]. There are also titanium-rich systems [10,11] based on heavy transition metals [12] and partly ordered systems, [13] (which have some of the characteristics of HEAs). The formation of superconducting lanthanide-containing HEAs was first reported as early as 2015 [14], but to date there is very little information on superconductivity in such systems. Recently, a paper about a superconducting partly ordered system containing heavy lanthanides [15] was published, but there is no information about superconductivity in a system containing lanthanides which crystallize in a bcc structure.

In this study, we aim to explore whether the incorporation of terbium (Tb), a lanthanide with strong magnetic properties, into a high-entropy alloy matrix can influence

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its superconducting behavior. Specifically, we investigate whether Tb can increase the critical temperature or the upper critical field or modify the electron–phonon coupling strength. These properties are critical for potential applications of superconductors in high-field environments. Given the limited data on lanthanide-containing HEAs, especially those crystallizing in a bcc structure, our goal is to determine whether such systems can be synthesized and to evaluate the impact of Tb on their superconducting parameters.

Our starting point was the NbTa matrix, but in the course of our research we found that combinations of tantalum with lanthanides are impossible to obtain in an arc furnace due to differences in melting points and thermal conductivity (see Supplementary Materials). Nevertheless, we were able to obtain a system with Tb, namely, $(VNb)_{0.67}(TiTbHf)_{0.33}$, and a corresponding lanthanide-free matrix with a formula $(VNb)_{0.753}(TiHf)_{0.247}$ to assess the effect of terbium on the superconducting properties of HEA.

2. Materials and Methods

Polycrystalline samples of $(VNb)_{0.67}(TiTbHf)_{0.33}$ and $(VNb)_{0.753}(TiHf)_{0.247}$ were synthesized by conventional arc melting in an atmosphere of Ti-treated Ar using elemental metal ingots of at least 99.9% purity. To remove residual, black terbium oxide from the surface of the pure terbium metal, the Tb ingot was polished using sandpaper. The melting process was repeated five times to ensure the homogeneity of the products. The mass loss was negligible (around 0.8% for Tb sample and below 0.2% for the reference). To enable structural and morphology studies, the samples were cut in half using a diamond circular saw and their surfaces polished.

The chemical composition and homogeneity of the samples were verified by energy-dispersive X-ray spectroscopy (EDXS) using a FESEM FEI Nova NanoSEM 230 scanning electron microscope equipped with an EDAX Genesis XM4 spectrometer on the surface of the cut and polished samples. The crystal structures of the as-obtained products were studied by powder X-ray diffraction (XRD) using a PANalytical X'pert Pro diffractometer with CuKα radiation. The experimental XRD patterns were analyzed using the Rietveld method implemented in the FULLPROF software [16].

The AC magnetic susceptibility of the alloys was studied in the temperature range of 1.8 K to 6 K and in magnetic fields up to 50 kOe using using a commercial Quantum Design MPMS-XL magnetometer. The heat capacity was measured from room temperature to 1.8 K and in fields up to 50 kOe using the Quantum Design PPMS and DynaCool platforms.

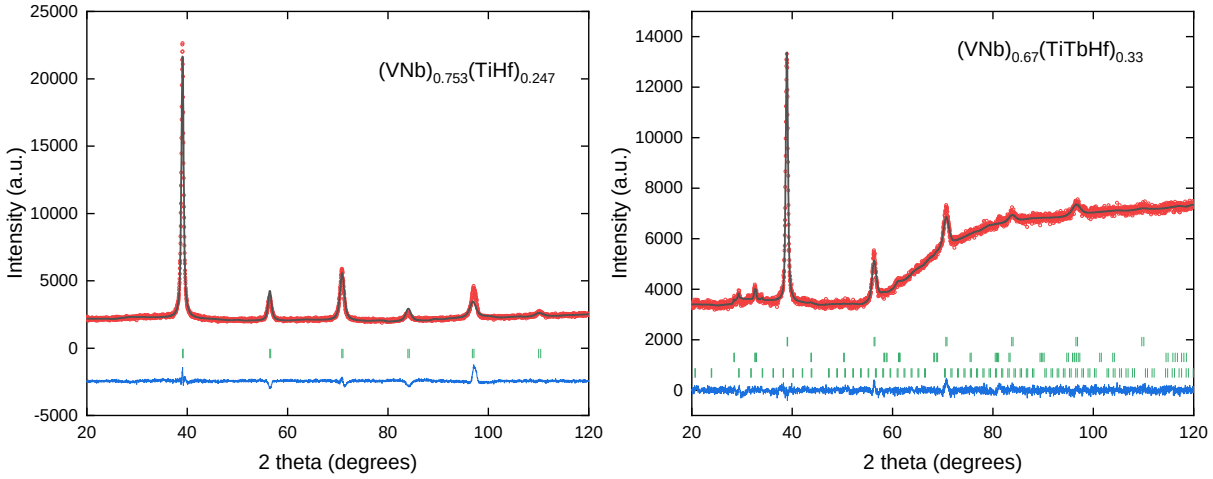
3. Results and Discussion

3.1. Crystal Structure

X-ray diffraction patterns of the obtained alloys are presented in Figure 1. The analysis of the experimental data by the Rietveld method showed that the $(VNb)_{0.753}(TiHf)_{0.247}$ alloy crystallizes in a body-centered cubic (bcc) structure; no traces of additional phases were detected. Reliability factors ($R_p = 2.81$, $R_{wp} = 4.23$) commonly used to estimate the quality of the refinement demonstrated the good quality of this fit. The lattice parameter of the matrix was found to be $a = 3.2568(1)$ Å.

The results of the XRD measurements indicate that there are three phases in the $(VNb)_{0.67}(TiTbHf)_{0.33}$ alloy: the expected structure of the bcc HEA, pure terbium precipitates with an amount of 2.9 wt.% estimated using Rietveld method, and a small amount (1.7 wt.%) of Tb_2O_3. The phases are listed in the order in which their Bragg reflections are presented (from top to bottom) in Figure 1b. The lattice parameter of the bcc phase was $a = 3.2642(1)$ Å. The slight increase in the crystal lattice parameter is most likely related to the incorporation of Tb atoms. It is evident that some of the terbium atoms did not dissolve in the main phase and formed precipitates of the pure element. In addition, a passiva-

tion layer of terbium oxide must have formed on the surface of these precipitates (as the measurement was conducted on the surface of the cut and polished specimen and any residual oxides should, during the melting, migrate to the outer surface of the alloy ingot). Reliability factors $R_p = 1.35$, $R_{wp} = 1.76$ demonstrated, again, the good quality of this fit.



(**a**) (**b**)

Figure 1. X-ray diffraction patterns and results of refinements for the (**a**) $(VNb)_{0.753}(TiHf)_{0.247}$ and (**b**) $(VNb)_{0.67}(TiTbHf)_{0.33}$ alloys. Red circles and gray lines represent the experimental points and the theoretical curves, respectively. Blue lines show a difference between the two, and green vertical dashes indicate positions of the Bragg reflections for the identified phases. In panel (**b**), the upper dashes correspond to bcc phase, the middle to pure Tb precipitates, and the bottom to Tb_2O_3.

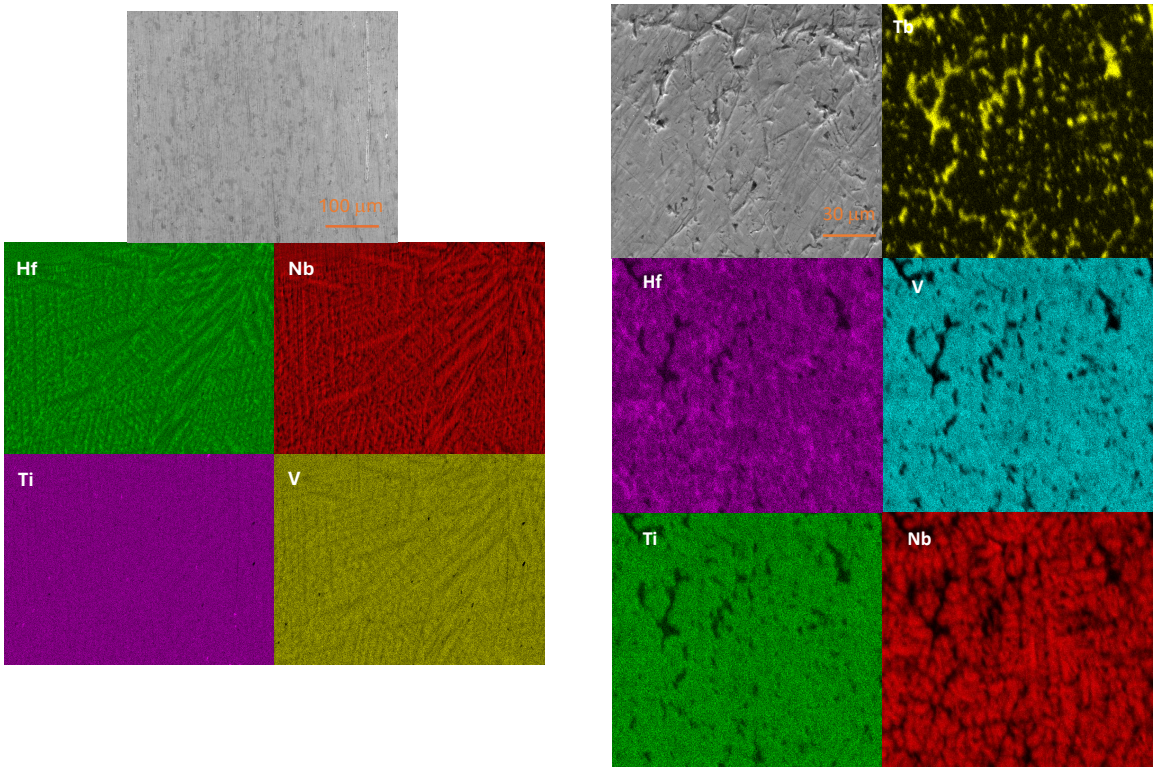
3.2. Morphology and Chemical Composition

The analysis of the EDXS maps of the$(VNb)_{0.753}(TiHf)_{0.247}$ alloy, presented in Figure 2a, showed that while the sample lacks pure metal precipitates or areas that could indicate the presence of intermetallic compounds, there is a slight segregation into a niobium-rich HEA phase and a niobium-poor phase, with the titanium and vanadium content appearing to fluctuate less (see Table 1). The two phases form a mosaic pattern, and their crystal lattice constants must be very close to each other, even almost identical, since their separation is not visible in XRD measurements.

Table 1. Averaged compositions over several repeated measurements of similar areas of the main phase in the $(VNb)_{0.67}(TiTbHf)_{0.33}$ alloy and two representative areas present in the $(VNb)_{0.753}(TiHf)_{0.247}$.

	Element	Ti	V	Tb	Hf	Nb
$(VNb)_{0.67}(TiTbHf)_{0.33}$	at.%	12.13 ± 0.63	37.92 ± 0.89	2.73 ± 0.43	14.13 ± 1.26	33.09 ± 1.81
$(VNb)_{0.753}(TiHf)_{0.247}$: Nb-rich	at.%	12.95 ± 0.65	38.32 ± 0.79	—	12.53 ± 0.66	36.19 ± 0.83
$(VNb)_{0.753}(TiHf)_{0.247}$: Nb-poor	at.%	13.09 ± 0.70	39.30 ± 1.2	—	15.06 ± 1.0	32.56 ± 2.0

The micrographs of the $(VNb)_{0.67}(TiTbHf)_{0.33}$ alloy, presented in Figure 2b, show clear precipitates of terbium (and possibly terbium oxide in the same area, although oxygen was not mapped). Moreover, the main phase has a slightly different morphology from the matrix. First, the mosaic structure is only visible at higher magnification. Second, segregation is visible in the case of niobium. Local disturbances in the stoichiometry of the other elements are also present but they are not significant. However, this is not reflected in the XRD measurements, where additional phases (beyond the three mentioned in the previous section) are not visible.



(**a**) (**b**)

Figure 2. SEM micrographs with EDXS elemental mapping superimposed on the sample surface of the (**a**) $(VNb)_{0.753}(TiHf)_{0.247}$ and (**b**) $(VNb)_{0.67}(TiTbHf)_{0.33}$ alloys.

3.3. Physical Properties

The results of the AC magnetic susceptibility measurements in various magnetic fields in the zero-field cooling (ZFC) regime are presented in Figures 3 and 4. In both alloys, the real and imaginary parts of the susceptibility confirm the presence of bulk, type II superconductivity. The incorporation of the terbium atoms into the matrix causes the reduction in T_c from about 5.2 K to 4.7 K. The critical temperatures were determined based on the position of the maximum in $\chi''(T)$ data.

Additionally, the degree of critical temperature change in relation to the induction of magnetic fields indicates that the sample containing terbium demonstrates a higher degree of decay in its superconductivity.

This happens because, as the temperature gets higher, magnetic flux lines and bulk shielding currents begin to penetrate the superconductor when the applied field exceeds the lower critical field. This destroys the ideal Meissner state and causes energy to be lost. When the flux lines and shielding currents fully penetrate the material, the losses reach a maximum. When the temperature reaches its critical value, the loss value goes to zero [17]. In the case of the$(VNb)_{0.753}(TiHf)_{0.247}$ sample, the superconducting state is still present in the field of 5 T, while the sample of $(VNb)_{0.67}(TiTbHf)_{0.33}$ becomes paramagnetic under the same conditions. No signals characteristic of any magnetic orders were detected. The main phase inhomogeneities seen in the SEM images must therefore have a negligible effect on the magnetic properties of the alloys.

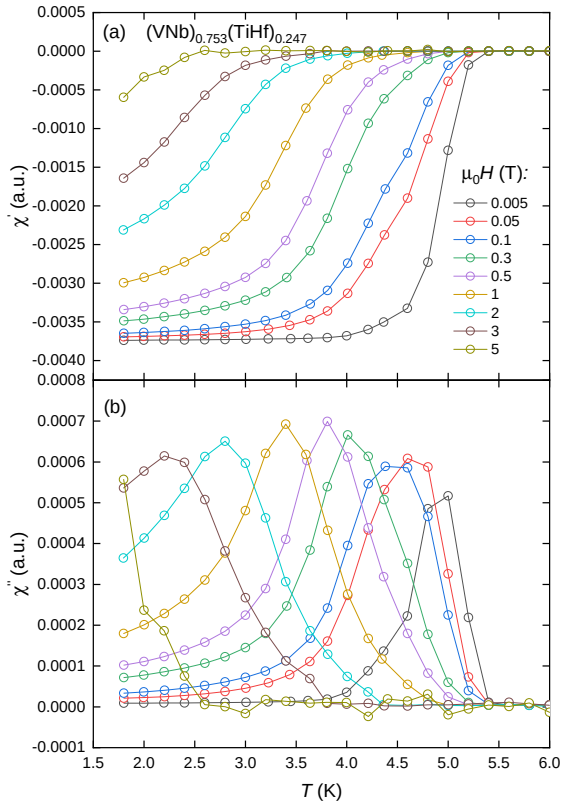


Figure 3. Real (**a**) and imaginary (**b**) parts of AC susceptibility measured for the $(VNb)_{0.753}(TiHf)_{0.247}$ alloy in the nominal applied magnetic fields $\mu_0 H$. Hollow circles represent the measured points, and solid lines serve as a visual guide.

Since the results of the magnetic measurements indicated the bulk nature of superconductivity, specific heat measurements were used for further analysis and to define the critical properties of the superconducting states. In the rest of this paper, we call the alloy $(VNb)_{0.753}(TiHf)_{0.247}$ 'matrix' and the $(VNb)_{0.67}(TiTbHf)_{0.33}$ alloy 'HEA-Tb'.

The results of the measurements of the specific heat of the two alloys at a nominal zero field and temperatures from RT to 1.8 K are presented in Figure 5a,c. The overall shape of the experimental C_p(T) curves indicate typical behavior of metal alloys. Namely, they follow the Dulong–Petit law and generally show no additional features except for the HEA-Tb alloy, where at about 220 K, an anomaly related to the ferromagnetic ordering of pure terbium can be observed [18]. The inserts of these two figures shows a low-temperature measurement range, revealing transitions to the superconducting state in both alloys studied. For the matrix, the transition has a typical λ shape and is quite sharp. T_c calculated by the equal entropy method is 5.2(1) K. In the case of HEA-Tb, the transition is broadened. Before the transition, a slight lifting of the measurement points is also observed. In addition, an anomaly related to the ordering of antiferromagnetic terbium oxide Tb_2O_3 is observed (about 2 K [19]). The critical temperature, determined by the same method, was 4.6(1) K. Normalized specific heat jump $\Delta C_p/\gamma T_c$ was found to be equal to 1.49 (matrix) and 1.05 (HEA-Tb). The latter is significantly smaller than the value predicted by the BSC theory. This may be caused by the effect of terbium doping or by the effect of magnetic phases on the height of the measurement points.

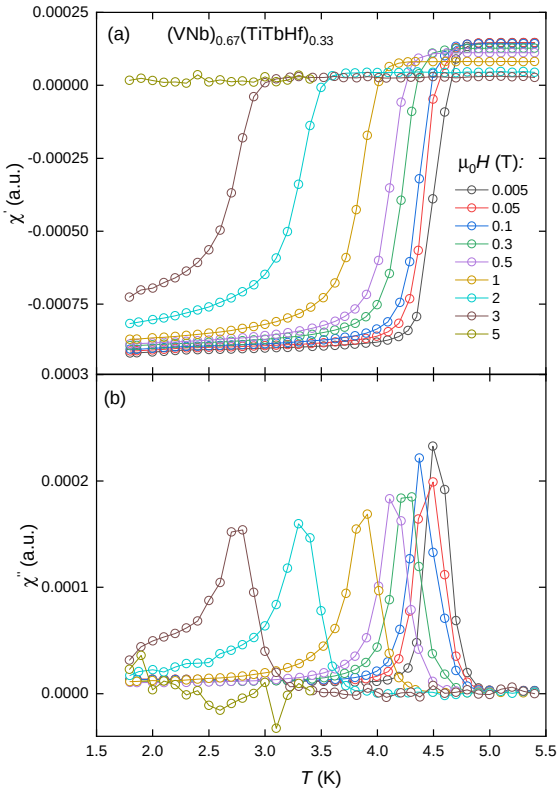


Figure 4. Real (**a**) and imaginary (**b**) parts of AC susceptibility measured for the $(VNb)_{0.67}(TiTbHf)_{0.33}$ alloy in the nominal applied magnetic fields $\mu_0 H$. Hollow circles represent the measured points, and solid lines serve as a visual guide.

At low temperatures the C_p(T) curve can be described as a sum of the electronic heat contribution (connected with γ coefficient) and the phonon heat contribution (with corresponding β and σ coefficients). The solid black curves represent the fits of Equation (1) to the experimental data.

$$C_P(T) = \gamma T + \beta T^3 + \sigma T^5, \quad (1)$$

The γ, β, and σ coefficients were found to be equal to 9.1(1) mJ K^{-2} mol^{-1}, 0.079(4) mJ K^{-4} mol^{-1}, and 5.6(4) $\times$ 10^{-4} mJ K^{-6} mol^{-1} for the matrix and 11.1(2) mJ K^{-2} mol^{-1}, 0.085(9) mJ K^{-4} mol^{-1}, and 6.4(1) $\times$ 10^{-4} mJ K^{-6} mol^{-1} for the HEA-Tb, accordingly.

Using the calculated value of the β coefficient and the Equation (2) (where R is a gas constant and r = 1), the Debye Θ_D temperatures were estimated for the two alloys as 291 K (matrix) and 284 K (HEA-Tb).

$$\beta = \frac{12}{5} r R \pi^4 \left(\frac{T}{\Theta_D}\right)^3, \quad (2)$$

The T_c and Θ_D values were used to calculate electron–phonon coupling using McMillan's Equation [20]

$$\lambda_{el-ph} = \frac{1.04 + \mu^* \ln\left(\frac{\Theta_D}{1.45 T_c}\right)}{(1 - 0.62\mu^*) \ln\left(\frac{\Theta_D}{1.45 T_c}\right) - 1.04}, \quad (3)$$

where μ^* is the Coulomb repulsion constant. Assuming $\mu^* = 0.125$ (a value commonly used for *d*-electron systems), we obtained $\lambda_{el-ph} = 0.68$ (matrix) and 0.62 (HEA-Tb), which places the alloys in weakly to intermediate coupling.

Using the relation:

$$\gamma = \frac{1}{3} \pi^2 k_B^2 N_A N(E_F) \quad (4)$$

where k_B is the Boltzmann constant and N_A is the Avogadro number. The experimental densities of states of conduction electrons at the Fermi level $N(E_F)$ were estimated as 3.9 states eV^{-1} f.u.$^{-1}$ (matrix) and 4.7 states eV^{-1} f.u.$^{-1}$ (HEA-Tb). The densities of non-interacting electrons $N(E_F)^*$ were calculated as 2.3 states eV^{-1} f.u.$^{-1}$ (matrix) and 2.9 states eV^{-1} f.u.$^{-1}$ (HEA-Tb) using the following relation [21]:

$$N(E_F)^* = \frac{N(E_F)}{1+\lambda_{el-ph}}. \quad (5)$$

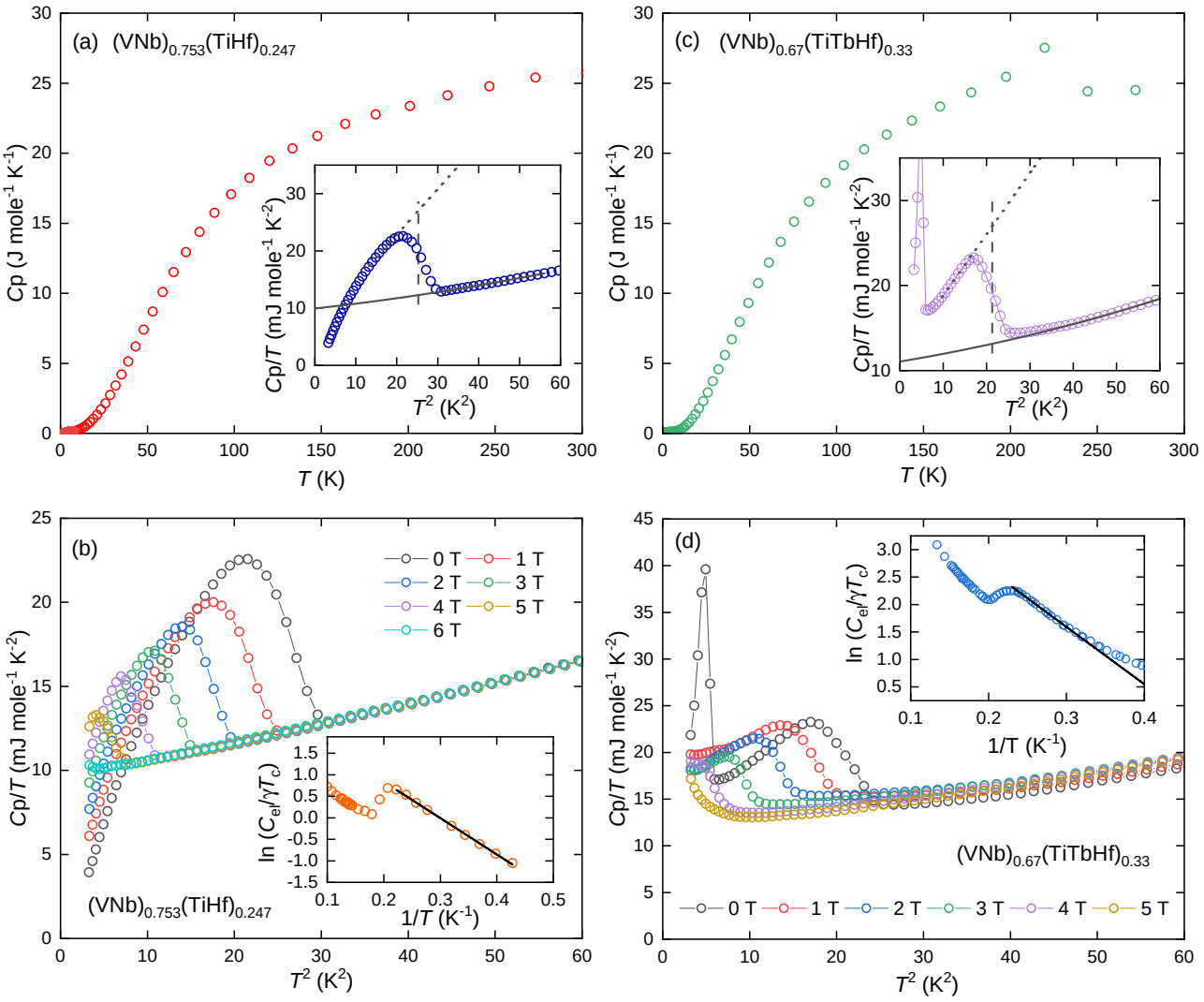


Figure 5. (**a**) Temperature variation of specific heat C_P of $(VNb)_{0.753}(TiHf)_{0.247}$; the insert shows $C_P/T(T^2)$ measured in nominal zero magnetic field, and the solid curve depicts fit of Equation (1) to the experimental data together with dashed lines indicating the estimated normalized jump and T_c. (**b**) C_P/T vs. T^2 measured in various external magnetic fields $\mu_0 H$ up to 6 T for $(VNb)_{0.753}(TiHf)_{0.247}$ sample; the inset shows $\ln(C_{el}/\gamma T_c)$ as a function of inverse temperature measured in a nominal zero field, and the straight solid line is a fit of Equation (6) to the experimental data. Panels (**c**,**d**) display analogous analysis for the $(VNb)_{0.67}(TiTbHf)_{0.33}$ alloy.

Then, by subtracting the major phonon contribution to the specific heat βT^3 from the total heat C_p, it is possible to estimate its pure electron contribution to the specific heat C_{el} and—from its change with temperature—the superconducting energy gap Δ_0. The inverse

temperature dependence of the electron heat is shown as $\ln(C_{el}/\gamma T_c)(1/T)$ in the insets to Figure 5b,d. The solid black regression lines depict the least-squares fits of Equation [22]:

$$C_{BCS}(T) = A\gamma T_c \exp\left(-\frac{\Delta_0}{k_B T}\right), \tag{6}$$

where A is a constant. By applying the above relation, we obtained $\Delta_0/k_B = 8.45$ K (matrix) and 10.41 K (HEA-Tb), which after normalizations yielded $2\Delta_0/k_B T_c = 3.25$ and 4.53, respectively. The latter deviates significantly from the BCS weak-coupling limit of 3.52, which is expected for a single, isotropic, superconducting gap. However, such elevated values are not uncommon in systems containing lanthanides and actinides [23]. This enhancement may point to strong electron–phonon coupling, consistent with the moderately high coupling constant ($\lambda_{el-ph} = 0.62$) derived for HEA-Tb using McMillan's formula.

The Pauli limiting fields $\mu_0 H_P = 9.4$ T (matrix) and 8.5 T (HEA-Tb) were then calculated from the following relation [24]:

$$\mu_0 H_P = 1.84 T_c. \tag{7}$$

Also, using the calculated values of the energy gap, the thermodynamic critical fields 0.11 T (matrix) and 0.14 T (HEA-Tb) were calculated from the relation [25]:

$$\mu_0 H_c(0) = \sqrt{\frac{3\gamma_V}{2\pi^2 \mu_0}} \frac{\Delta_0}{k_B}. \tag{8}$$

where γ_V is the volumetric Sommerfeld coefficient.

To estimate the values of the upper critical field $\mu_0 H_{c2}(0)$, we studied the behavior of the anomalies in specific heat in different magnetic fields (see Figure 5b,d). The critical temperature values were used to construct the phase diagram (Figure 6). To make the estimation accurate, the Werthamer–Helfand–Hohenberg (WHH) [26–28] model for BCS superconductors in the dirty limit was used:

$$\ln\frac{1}{t} = \left(\frac{1}{2} + \frac{i\lambda_{SO}}{4\gamma}\right)\psi\left(\frac{1}{2} + \frac{\overline{h} + \frac{1}{2}\lambda_{SO} + i\gamma}{2t}\right) + \left(\frac{1}{2} - \frac{i\lambda_{SO}}{4\gamma}\right)\psi\left(\frac{1}{2} + \frac{\overline{h} + \frac{1}{2}\lambda_{SO} - i\gamma}{2t}\right) - \psi\left(\frac{1}{2}\right) \tag{9}$$

where α_M is the Maki constant, λ_{SO} is the spin–orbit scattering constant, $\gamma \equiv \sqrt{(\alpha_M \overline{h})^2 - (\frac{1}{2}\lambda_{SO})^2}$, $\overline{h} = \frac{4}{\pi^2}\frac{H_{c2}}{-dH_{c2}/dt}$, $t = \frac{T}{T_C}$, and ψ is a digamma function. The model includes the spin-paramagnetic effect and allowed the estimation of $\mu_0 H_{c2}(0)$ (6.8 T for the matrix and 6.1 T for HEA-Tb), the orbital upper critical field $\mu_0 H_{c2}^{orb}$ (7.9(1) T for the matrix and 7.3(1) T for HEA-Tb), and α_M (1.18 for the matrix and 1.20 for HEA-Tb). Lowering the upper critical field by doping with lanthanide (terbium) therefore has the opposite effect to raising it when doped with scandium, a rare earth element, as presented in [29]. It is worth noting that XRD and SEM-EDXS analysis revealed the presence of secondary phases, specifically, Tb precipitates and Tb_2O_3. These phases likely contribute to the observed reduction in the superconducting volume fraction in the HEA-Tb sample. As these phases do not participate in the superconducting state, they may act as non-superconducting inclusions, thereby reducing the effective superconducting volume. While such inclusions can enhance flux pinning in some systems by acting as pinning centers, our results suggest that this effect is weak or even negligible. The lower upper critical field and broader superconducting transition in the HEA-Tb alloy support this interpretation.

In Figure 5d and in the inset of Figure 5c, it can be seen that there is some lifting of the measurement points before the main transition. Most likely, this is related to the Tb concentration gradient at the interphase between the alloy and pure Tb, similar to what

was reported in the case of thorium-rich and thorium-poor phases in another HEA alloy superconductor [30].

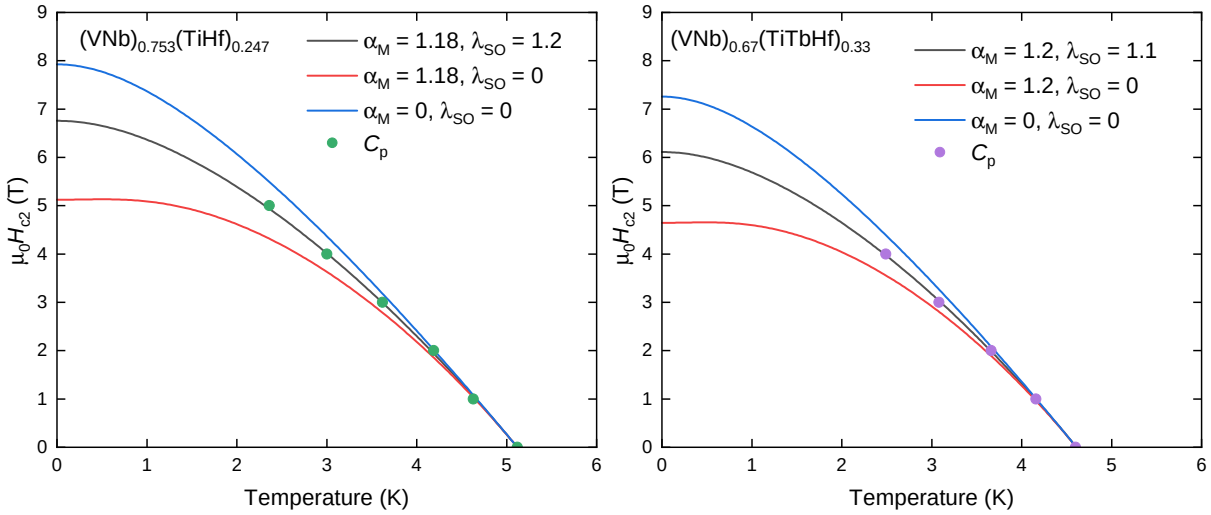


Figure 6. Upper critical fields of (**a**) $(VNb)_{0.753}(TiHf)_{0.247}$ and (**b**) $(VNb)_{0.67}(TiTbHf)_{0.33}$ as a function of temperature, derived from specific heat data. The lines depict the simulated WHH curves (Equation (9)).

The non-zero values of the spin–orbit scattering parameter λ_{SO} obtained from the WHH fits indicate that spin–orbit coupling plays a non-negligible role in the superconducting behavior of both alloys. This is particularly relevant in multicomponent systems, where the presence of heavy elements can enhance spin–orbit interactions. The extracted λ_{SO} values (1.2 for the matrix and 1.1 for HEA-Tb) suggest moderate spin–orbit scattering, which may influence the robustness of the superconducting state in the case of applied magnetic fields. These values are comparable to those reported for the $(NbTi)_{0.67}(MoHfV)_{0.33}$ superconductor, which also exhibits comparable $\mu_0 H_{c2}(0) = 6.8(1)$ T [31]. In contrast, HEAs with significantly higher upper critical fields ($\mu_0 H_{c2}(0) > 12$ T), such as $Ti_{0.5}(ZrNbHfTa)_{0.5}$ and $Ti_{0.5}(VNbHfTa)_{0.5}$, show λ_{SO} values exceeding 3 [11]. This indicates a possible correlation between strong spin–orbit scattering and enhanced critical field in HEA superconductors.

In the next step, the Ginzburg–Landau parameters were estimated using the $\mu H_{c2}(0)$, $\mu_0 H_{c2}^{orb}$ and $\mu_0 H_c$. The coherence lengths at 0 K $\xi_{GL}(0)$ [32] are as follows:

$$\xi_{GL}(0) = \frac{\phi_0}{2\pi\mu_0 H_{c2}^{orb}(0)}, \tag{10}$$

and were found to be 6.4 nm (matrix) and 6.7 nm (HEA-Tb). The values of the Ginzburg–Landau penetration depth at 0 K $\lambda_{GL}(0)$ were determined as follows:

$$\lambda_{GL}(0) = \sqrt{\frac{\phi_0 \mu_0 H_{c2}^{orb}(0)}{4\pi\mu_0 H_c(0)^2}}, \tag{11}$$

and were found to be 320 nm (matrix) and 246 nm (HEA-Tb). Finally, the Ginzburg–Landau parameters κ_{GL} were defined as follows:

$$\kappa_{GL} = \frac{\lambda_{GL}}{\xi_{GL}}, \tag{12}$$

and were calculated as equal to 50 for matrix and 37 for HEA-Tb. Based on κ_{GL} and the thermodynamic critical field, the lower critical fields $\mu_0 H_{c1}(0)$ were calculated using the following relation:

$$\mu_0 H_{c1}(0) = \frac{\mu_0 H_c(0)}{\sqrt{2}\kappa_{GL}} ln(\kappa_{GL}), \quad (13)$$

as equal to 0.0063 T (matrix) and 0.0097 T (HEA-Tb).

All the derived parameters describing the physical properties of both studied alloys are presented in Table 2.

Table 2. Experimentally determined basic characteristic parameters of normal and superconducting states in $(VNb)_{0.753}(TiHf)_{0.247}$ and $(VNb)_{0.67}(TiTbHf)_{0.33}$.

Parameter	$(VNb)_{0.753}(TiHf)_{0.247}$	$(VNb)_{0.67}(TiTbHf)_{0.33}$
T_c	5.2(1) K	4.6(1) K
γ	9.1(1) mJ K^{-2} mol^{-1}	11.1(2) mJ K^{-2} mol^{-1}
β	0.079(4) mJ K^{-4} mol^{-1}	0.085(9) mJ K^{-4} mol^{-1}
σ	5.6(4) $\times$ 10^{-4} mJ K^{-6} mol^{-1}	6.4(1) $\times$ 10^{-4} mJ K^{-6} mol^{-1}
Θ_D	291(5) K	284(10) K
$\Delta C_p/\gamma T_c$	1.49	1.05
λ_{el-ph}	0.68(1)	0.62(1)
$N(E_F)$	3.9(1) states eV^{-1} f.u.$^{-1}$	4.7(1) states eV^{-1} f.u.$^{-1}$
$N(E_F)^*$	2.3(1) states eV^{-1} f.u.$^{-1}$	2.9(1) states eV^{-1} f.u.$^{-1}$
$\mu_0 H_c(0)$	0.11 T	0.14 T
$2\Delta_0/(k_B T_c)$	3.25	4.53
$\mu_0 H_{c1}(0)$	0.0063 T	0.0097 T
$\mu_0 H_{c2}(0)$	6.8 T	6.1 T
$\mu_0 H_{c2}^{orb}$	7.9(1) T	7.3(1) T
$\mu_0 H_P$	9.4(1) T	8.5(1) T
α_M	1.18	1.20
$\xi_{GL}(0)$	6.4 nm	6.7 nm
$\lambda_{GL}(0)$	320 nm	246 nm
κ_{GL}	50	37

4. Conclusions

The synthesis, structural, and physical properties of the high-entropy alloy system $(VNb)_{0.67}(TiTbHf)_{0.33}$ and the corresponding Tb-free matrix $(VNb)_{0.753}(TiHf)_{0.247}$ were reported and discussed. Refinement of the XRD data showed that both alloys crystallized in body-centered cubic structures, and in the case of the former alloy, some inhomogeneity of pure Tb and its oxide was detected. The results were confirmed by SEM-EDXS observations, which additionally showed some minor inhomogeneities in the main phases of both alloys. Measurements of magnetic susceptibility and specific heat have shown that terbium doping does not improve the critical parameters of the superconducting state of the bcc HEA alloy. However, there is no clear indication whether it is caused by the innate magnetic properties of terbium or an effect related to the electronic structure; therefore, further studies are required, possibly using other lanthanides and/or actinides.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/ma18122747/s1, Table S1: Selected physical properties of metals used in attempts to synthesize HEA alloys with lanthanides.

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